

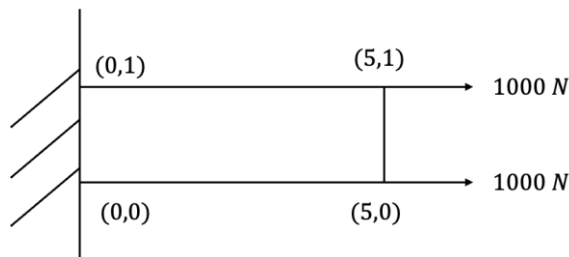
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Date: 2026-02-27

Intro to Finite Element Method (CIV E – 665)
Assignment No. 6

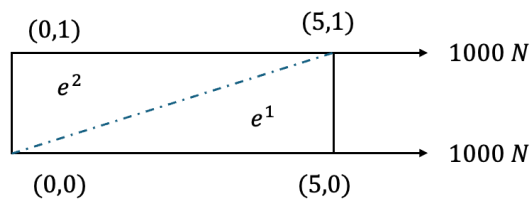


Question 1 (70 Points)

Consider the following structure. Solve for the nodal displacements assuming a plane stress condition (with $t = 0.2$ m) and that the elastic modulus and Poisson's ratio are 100 GPa and 0.3 respectively. Assume that the structure has no body forces.



Solve for the **nodal displacements** using **two linear triangular elements** as shown below:



NB: The coordinates are expressed in m.

For 2D plane elasticity the weakform in matrix notation is shown at Equation 1.

$$\underbrace{\left[\int_{\Omega} [B]^T [C] [B] t dx dy \right]}_{\text{Stiffness}} \underbrace{\{u\}}_{\substack{\text{Nodal} \\ \text{Displacements}}} = \underbrace{\left\{ \int_{\Omega} [\psi]^T \begin{Bmatrix} f_x \\ f_y \end{Bmatrix} t dx dy \right\}}_{\text{Body Force}} + \underbrace{\oint_{\Gamma} [\psi]^T \begin{Bmatrix} t_x \\ t_y \end{Bmatrix} t dS}_{\text{Traction: Natural BC}} \quad (1)$$

$[B]$ is the strain displacement matrix, consists of the derivatives of the shape functions ψ_n . The size of B depends on the type of element, which can be expressed as Equation 2.

$$B = \begin{bmatrix} \frac{\partial \psi_1}{\partial x} & 0 & \frac{\partial \psi_2}{\partial x} & 0 & \frac{\partial \psi_3}{\partial x} & \dots & \frac{\partial \psi_n}{\partial x} & 0 \\ 0 & \frac{\partial \psi_1}{\partial y} & 0 & \frac{\partial \psi_2}{\partial y} & 0 & \frac{\partial \psi_3}{\partial y} & \dots & \frac{\partial \psi_n}{\partial y} \\ \frac{\partial \psi_1}{\partial y} & \frac{\partial \psi_1}{\partial x} & \frac{\partial \psi_2}{\partial y} & \frac{\partial \psi_2}{\partial x} & \frac{\partial \psi_3}{\partial y} & \dots & \frac{\partial \psi_n}{\partial y} & \frac{\partial \psi_n}{\partial x} \end{bmatrix}_{3 \times 2n} \quad (2)$$

For the triangular element, B only has three shape functions, as shown in Equation 3.

$$B = \begin{bmatrix} \frac{\partial \psi_1}{\partial x} & 0 & \frac{\partial \psi_2}{\partial x} & 0 & \frac{\partial \psi_3}{\partial x} & 0 \\ 0 & \frac{\partial \psi_1}{\partial y} & 0 & \frac{\partial \psi_2}{\partial y} & 0 & \frac{\partial \psi_3}{\partial y} \\ \frac{\partial \psi_1}{\partial y} & \frac{\partial \psi_1}{\partial x} & \frac{\partial \psi_2}{\partial y} & \frac{\partial \psi_2}{\partial x} & \frac{\partial \psi_3}{\partial y} & \frac{\partial \psi_3}{\partial x} \end{bmatrix}_{3 \times 6} \quad (3)$$

For linear triangular elements, in general, ψ_j can be written as Equation 4.

$$\psi_j = \alpha_j + \beta_j x + \gamma_j y \quad (4)$$

Number the element 1 and element 2 as the red numbering shown as the Figure 1.

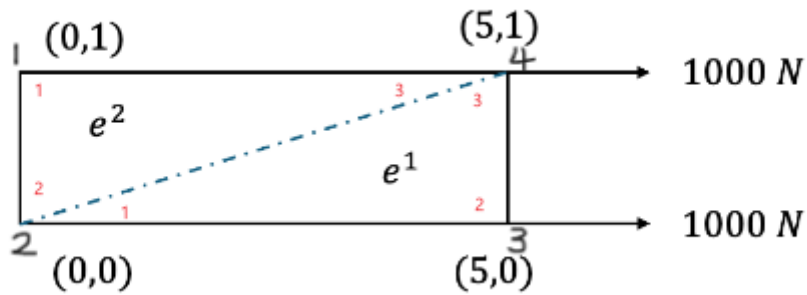


Figure 1: Numbering the elements

$\alpha\beta\gamma$ are constants evaluated from the coordinates for the coefficient matrix.

$$[\psi] = \begin{bmatrix} \alpha_1 & \alpha_2 & \alpha_3 \\ \beta_1 & \beta_2 & \beta_3 \\ \gamma_1 & \gamma_2 & \gamma_3 \end{bmatrix} = \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{bmatrix}^{-1}$$

Based on our numbering system, for ψ^1 ,

$$[\psi^1] = \begin{bmatrix} \alpha_1 & \alpha_2 & \alpha_3 \\ \beta_1 & \beta_2 & \beta_3 \\ \gamma_1 & \gamma_2 & \gamma_3 \end{bmatrix} = \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{bmatrix}^{-1}$$

Using Matlab get the ψ^1 and ψ^2

```
E1 = [1,0,0;1,5,0;1,5,1]
E2 = [1,0,1;1,0,0;1,5,1]
psi1 = inv(E1)
psi2 = inv(E2)
```

Output:

```
E1 =
     1     0     0
     1     5     0
     1     5     1
E2 =
     1     0     1
     1     0     0
     1     5     1
psi1 =
     1.0000     0     0
    -0.2000     0.2000     0
         0    -1.0000     1.0000
psi2 =
         0     1.0000     0
    -0.2000         0     0.2000
     1.0000    -1.0000         0
```

Substitute ψ in to Equation 3.

$$B^1 = \begin{bmatrix} -0.2 & 0 & 0.2 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1.0 & 0 & 1 \\ 0 & -0.2 & -1.0 & 0.2 & 1 & 0 \end{bmatrix}_{3 \times 6} \quad (5)$$

$$B^2 = \begin{bmatrix} 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0.2 & 0 & 0 \\ 0 & 1 & 0.2 & 0 & 0 & -1 \end{bmatrix}_{3 \times 6}$$

The $[C]$ consists of material constants.

$$C = \begin{bmatrix} c_{11} & c_{12} & 0 \\ c_{12} & c_{22} & 0 \\ 0 & 0 & c_{66} \end{bmatrix}_{3 \times 3}$$

For plain strain

$$c_{11} = c_{22} = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)}$$

$$c_{12} = \frac{E\nu}{(1+\nu)(1-2\nu)}$$

$$c_{66} = \frac{E}{2(1+\nu)}$$

Substitute the given data and use Matlab to solve it.

C1 =

```
1.0e+11 *  
  
1.3462    0.5769         0  
0.5769    1.3462         0  
         0         0    0.3846
```

Substitute B, C, E, t into Equation 6 and use Matlab solve it.

$$[K^e] = \left[[B^e]^T [C^e] [B^e] \int_{\Omega} \overbrace{t^e dx dy}^{dV} \right] \quad (6)$$

K1 =

```
1.0e+10 *  
         0         0         0         0         0         0  
         0         0         0         0         0         0  
         0         0    0.2692         0   -1.3462    0.5769  
         0         0         0    0.0769    0.3846   -0.3846  
         0         0   -1.3462    0.3846    8.6538   -4.8077  
         0         0    0.5769   -0.3846   -4.8077    8.6538
```

K2 =

```
1.0e+10 *  
6.7308         0         0    0.5769   -6.7308         0  
         0    1.9231    0.3846         0         0   -1.9231  
         0    0.3846    0.0769         0         0   -0.3846  
0.5769         0         0    0.2692   -0.5769         0  
-6.7308         0         0   -0.5769    6.7308         0  
         0   -1.9231   -0.3846         0         0    1.9231
```

Assemble element wise $[K]$, in our numbering case, for element 1

$$\begin{bmatrix} K_{11}^1 & K_{12}^1 & K_{13}^1 & K_{14}^1 & K_{15}^1 & K_{16}^1 \\ K_{21}^1 & K_{22}^1 & K_{23}^1 & K_{24}^1 & K_{25}^1 & K_{26}^1 \\ K_{31}^1 & K_{32}^1 & K_{33}^1 & K_{34}^1 & K_{35}^1 & K_{36}^1 \\ K_{41}^1 & K_{42}^1 & K_{43}^1 & K_{44}^1 & K_{45}^1 & K_{46}^1 \\ K_{51}^1 & K_{52}^1 & K_{53}^1 & K_{54}^1 & K_{55}^1 & K_{56}^1 \\ K_{61}^1 & K_{62}^1 & K_{63}^1 & K_{64}^1 & K_{65}^1 & K_{66}^1 \end{bmatrix} \begin{Bmatrix} u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \\ u_{4x} \\ u_{4y} \end{Bmatrix} = \begin{Bmatrix} f_{1x}^1 \\ f_{1y}^1 \\ f_{2x}^1 \\ f_{2y}^1 \\ f_{3x}^1 \\ f_{3y}^1 \end{Bmatrix}$$

for element 2

$$\begin{bmatrix} K_{11}^2 & K_{12}^2 & K_{13}^2 & K_{14}^2 & K_{15}^2 & K_{16}^2 \\ K_{21}^2 & K_{22}^2 & K_{23}^2 & K_{24}^2 & K_{25}^2 & K_{26}^2 \\ K_{31}^2 & K_{32}^2 & K_{33}^2 & K_{34}^2 & K_{35}^2 & K_{36}^2 \\ K_{41}^2 & K_{42}^2 & K_{43}^2 & K_{44}^2 & K_{45}^2 & K_{46}^2 \\ K_{51}^2 & K_{52}^2 & K_{53}^2 & K_{54}^2 & K_{55}^2 & K_{56}^2 \\ K_{61}^2 & K_{62}^2 & K_{63}^2 & K_{64}^2 & K_{65}^2 & K_{66}^2 \end{bmatrix} \begin{Bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{4x} \\ u_{4y} \end{Bmatrix} = \begin{Bmatrix} f_{1x}^2 \\ f_{1y}^2 \\ f_{2x}^2 \\ f_{2y}^2 \\ f_{3x}^2 \\ f_{3y}^2 \end{Bmatrix}$$

Notice that the element 1 and the element 2 have the common nodes u2 and u4. assemble them and apply the B.C condition.

$$\begin{bmatrix} K_{11}^2 & K_{12}^2 & K_{13}^2 & K_{14}^2 & 0 & 0 & K_{15}^2 & K_{16}^2 \\ K_{21}^2 & K_{22}^2 & K_{23}^2 & K_{24}^2 & 0 & 0 & K_{25}^2 & K_{26}^2 \\ K_{31}^2 & K_{32}^2 & K_{11}^1 + K_{33}^2 & K_{12}^1 + K_{34}^2 & K_{13}^1 & K_{14}^1 & K_{15}^1 + K_{35}^2 & K_{16}^1 + K_{36}^2 \\ K_{41}^2 & K_{42}^2 & K_{21}^1 + K_{43}^2 & K_{22}^1 + K_{44}^2 & K_{23}^1 & K_{24}^1 & K_{25}^1 + K_{45}^2 & K_{26}^1 + K_{46}^2 \\ 0 & 0 & K_{31}^1 & K_{32}^1 & K_{33}^1 & K_{34}^1 & K_{35}^1 & K_{36}^1 \\ 0 & 0 & K_{41}^1 & K_{42}^1 & K_{43}^1 & K_{44}^1 & K_{45}^1 & K_{46}^1 \\ K_{51}^2 & K_{52}^2 & K_{51}^1 + K_{53}^2 & K_{52}^1 + K_{54}^2 & K_{53}^1 & K_{54}^1 & K_{55}^1 + K_{55}^2 & K_{56}^1 + K_{56}^2 \\ K_{61}^2 & K_{62}^2 & K_{61}^1 + K_{63}^2 & K_{62}^1 + K_{64}^2 & K_{63}^1 & K_{64}^1 & K_{65}^1 + K_{65}^2 & K_{66}^1 + K_{66}^2 \end{bmatrix} \begin{Bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \\ u_{4x} \\ u_{4y} \end{Bmatrix} = \begin{Bmatrix} f_{2x}^2 \\ f_{2y}^2 \\ f_{1x}^1 + f_{2x}^2 \\ f_{1y}^1 + f_{2y}^2 \\ f_{2x}^1 \\ f_{2y}^1 \\ f_{3x}^1 + f_{3x}^2 \\ f_{3y}^1 + f_{3y}^2 \end{Bmatrix}$$

Due to node 1 and node 2 are fixed, so we can use 0 to simplify this matrix.

$$1.0 \times 10^{10} \begin{bmatrix} 2.1429 & -0.7143 & -1.9231 & 0.3297 \\ -0.7143 & 5.5714 & 0.3846 & -5.4945 \\ -1.9231 & 0.3846 & 2.1429 & 0 \\ 0.3297 & -5.4945 & 0 & 5.5714 \end{bmatrix} \begin{Bmatrix} u_{3x} \\ u_{3y} \\ u_{4x} \\ u_{4y} \end{Bmatrix} = \begin{Bmatrix} 1000 \\ 0 \\ 1000 \\ 0 + 0 \end{Bmatrix}$$

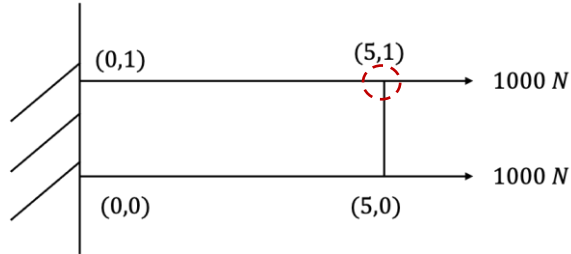
Then use Matlab to solve it.

```
Nodal Displacements [u3x; u3y; u4x; u4y] (in meters):
1.0e-06 *
0.4816
0.0387
0.4719
0.0097
```

Question 2 (30 Points)

Repeat **Question 1** in ABAQUS. You may follow the steps shown in the video ([link](#)).

- I. **Compare** the nodal displacement u_x of the highlighted node from **Question 1** with the corresponding result from **Question 2**



- II. **Show a screenshot** of ε_{xx} .

Hint to get exactly two elements use a global mesh size of 5

Follow the simulation tutorial, the ux results I got is shown as the Figure 2

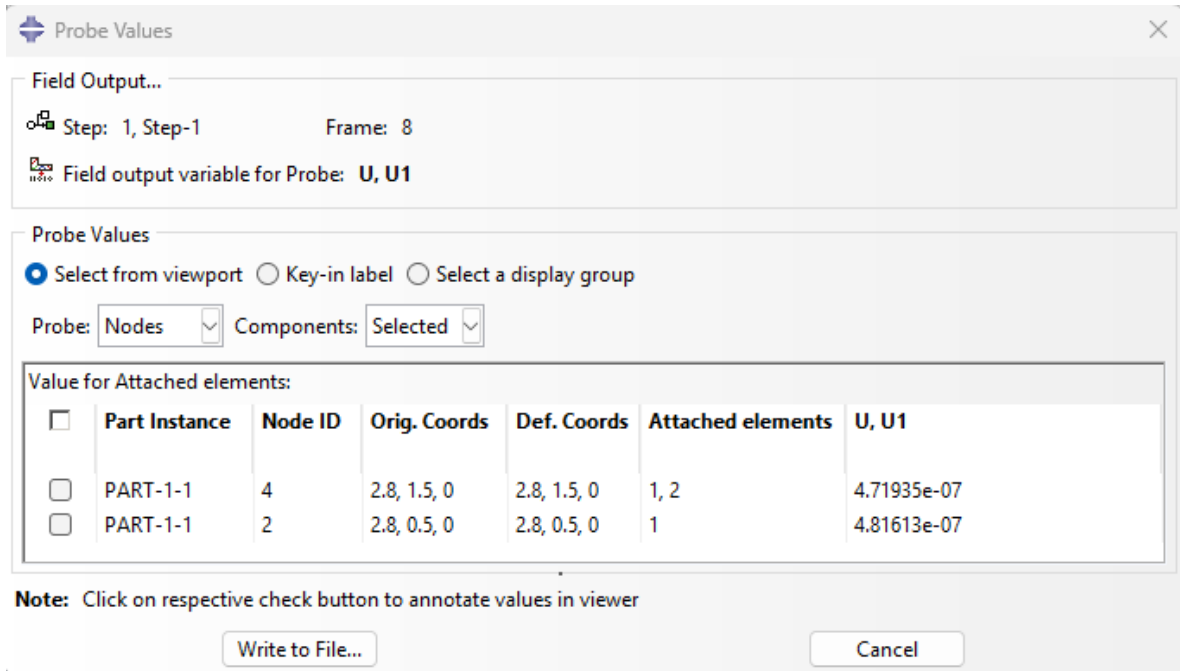


Figure 2: ux from abaqus

The node 4 in abaqus is my node 3, the node 2 is my node 4. I got same values.

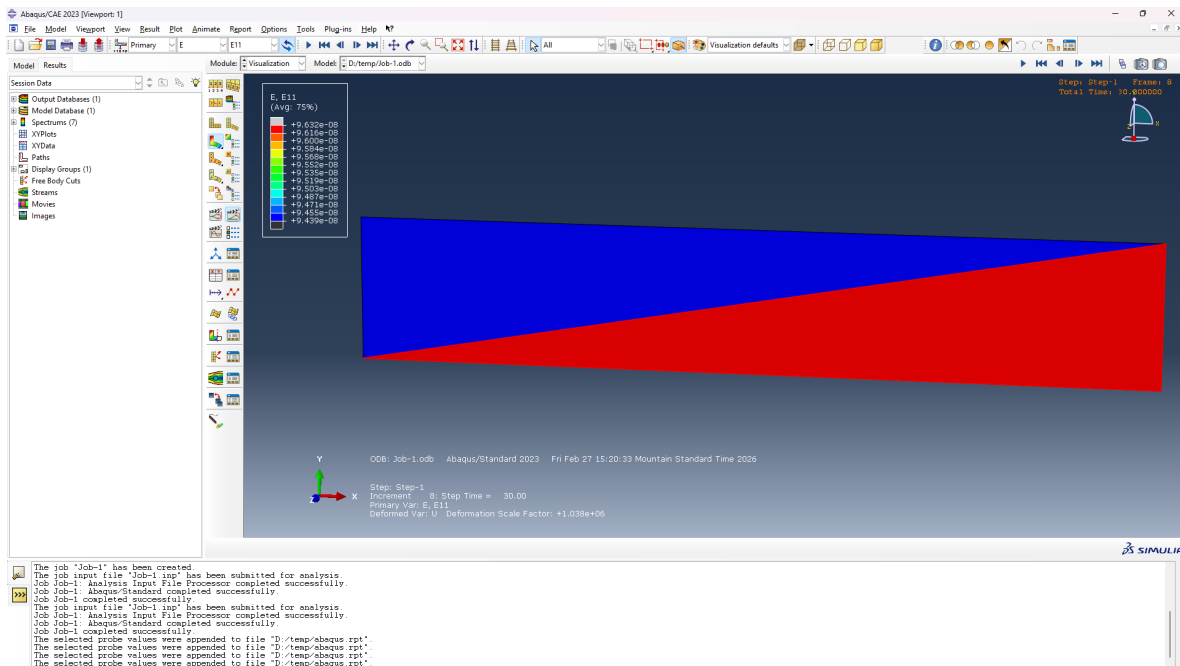


Figure 3: The screenshot of ϵ_{xx}